

# **cnavier**

## Numerical Methodology and Scientific User Manual

cnavier project

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### **Abstract**

**cnavier** is a compact two-dimensional incompressible Navier–Stokes solver written in C, with optional OpenMP and CUDA acceleration. The code advances the scalar vorticity equation and reconstructs a divergence-free velocity field from a stream function. Spatial derivatives are represented by finite-difference matrices of selectable interior order (second, fourth, or sixth), assembled into two-dimensional sparse operators through Kronecker products and applied in compressed sparse row (CSR) form. Time integration is explicit, using either forward Euler or classical fourth-order Runge–Kutta (RK4). The stream-function Poisson equation may be solved by Gauss–Seidel, successive over-relaxation (SOR), or a direct discrete-sine-transform solver implemented with FFTW3 on the CPU and cuFFT on the GPU.

This document describes the mathematical model, the exact discretizations present in the source code, solver sequencing, boundary treatment, stability considerations, verification and validation strategy, parallel backends, usage, and current numerical limitations. It is intended to be both a scientific description of the methodology and a practical reference for users and developers.

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# 1 Introduction

The objective of `cnavier` is to provide a small, inspectable computational fluid dynamics (CFD) code for two-dimensional incompressible viscous flow. Its reference problem is the square lid-driven cavity: three stationary no-slip walls and one tangentially moving wall. Despite the small code base, the project contains several numerical alternatives that are useful for study and benchmarking: multiple finite-difference stencil widths, two explicit time integrators, three elliptic solvers, a sparse-operator formulation, shared-memory parallelism, and an optional CUDA implementation of the complete time loop.

The code uses a *vorticity-streamfunction* formulation rather than the primitive-variable  $(u, v, p)$  equations. This choice removes pressure from the evolution system and replaces the pressure-velocity coupling problem by a scalar Poisson equation. In two dimensions, the streamfunction definition also gives a particularly strong discrete incompressibility property: when the discrete  $x$ - and  $y$ -derivative operators commute, the recovered velocity has zero discrete divergence to round-off.

The solver is designed primarily as an educational and experimental code rather than as a general-purpose production CFD package. The current implementation assumes a rectangular Cartesian domain, a Cartesian computational array, constant Reynolds number, explicit time stepping, and Dirichlet wall velocities. These constraints simplify the formulation sufficiently that each numerical operation can be followed directly from the governing equations to its C or CUDA implementation.

## 1.1 Repository organization

The main numerical components are:

- `src/main.c`: run configuration, command line, operator construction, time loop, diagnostics, and output;
- `src/backend.c`: one interface over the CPU and CUDA solvers, and the choice between them;
- `src/finitediff.c`: one-dimensional first- and second-derivative finite-difference matrices, with walls or periodic, and CSR conversion;
- `src/linearalg.c`: dense storage utilities, CSR sparse matrices, sparse matrix-vector multiplication, and Kronecker products;
- `src/fluiddyn.c`: wall conditions, vorticity transport, Euler and RK4 integration, recovery of the velocity from the streamfunction, and full timesteps;
- `src/poisson.c`: Gauss-Seidel, SOR, and FFTW/DST-I Poisson solvers, and the periodic FFT solver;
- `src/cudasolver.cu`: CUDA implementations of the time loop, sparse operators, elliptic solvers, and diagnostics;
- `src/threads.c`: the default number of OpenMP threads (one per physical core);
- `src/diagnostics.c`: energy, enstrophy and palinstrophy, and on periodic grids spectra and spectral fluxes;
- `src/utils.c`: VTK output, centerline data for comparison with Ghia et al., and the available-memory check;
- `tests/test_solver.c`: CPU verification and CPU-GPU parity tests.

## 2 Governing Equations

### 2.1 Nondimensional incompressible Navier–Stokes equations

For an incompressible Newtonian fluid of constant density and viscosity, the nondimensional two-dimensional Navier–Stokes equations are

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \frac{1}{\text{Re}} \nabla^2 \mathbf{u}, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

where

$$\mathbf{u} = (u, v)^T$$

is the velocity,  $p$  is the kinematic pressure, and

$$\text{Re} = \frac{U_{\text{ref}} L_{\text{ref}}}{\nu}$$

is the Reynolds number.

For the default lid-driven cavity,  $L_{\text{ref}} = 1$  and the lid speed is  $U_{\text{ref}} = 1$ , so the default nondimensional boundary data are

$$\begin{aligned} (u, v) &= (1, 0) && \text{on the moving lid,} \\ (u, v) &= (0, 0) && \text{on the other three walls.} \end{aligned}$$

### 2.2 Vorticity–streamfunction formulation

Define the scalar vorticity

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \quad (3)$$

Taking the curl of eq. (1) eliminates pressure and yields the vorticity transport equation

$$\frac{\partial \omega}{\partial t} + u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} = \frac{1}{\text{Re}} \left( \frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right). \quad (4)$$

Introduce a streamfunction  $\psi$  by

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (5)$$

Substitution into eq. (3) gives

$$\nabla^2 \psi = -\omega. \quad (6)$$

Thus a timestep consists conceptually of transporting  $\omega$ , solving the elliptic problem eq. (6), and differentiating  $\psi$  to obtain the velocity.

### 2.3 Why incompressibility is naturally enforced

At the continuous level, eq. (5) gives

$$\nabla \cdot \mathbf{u} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} = 0.$$

The code mirrors this property discretely. Let  $D_x$  and  $D_y$  denote the two-dimensional derivative matrices. The recovered velocity is

$$\mathbf{u}_h = \begin{bmatrix} D_y \psi \\ -D_x \psi \end{bmatrix}. \quad (7)$$

The discrete divergence is therefore

$$D_x D_y \psi - D_y D_x \psi. \quad (8)$$

Because the operators are built as Kronecker products acting in separate coordinate directions, they commute algebraically under the implemented layout. Consequently, the divergence diagnostic is expected to be near machine precision, apart from floating-point effects and details of the boundary rows. This behavior is explicitly tested by the project.

### 3 Computational Grid and Data Layout

#### 3.1 Domain and spacing

The physical domain is specified by  $L_x$  and  $L_y$  and the computational array by  $n_x \times n_y$ . The implementation defines

$$\Delta x = \frac{L_x}{n_x - 1}, \quad \Delta y = \frac{L_y}{n_y - 1}, \quad (9)$$

$n_x$  and  $n_y$  are independent; each may range from 8 through 16384.

Fields are stored as contiguous row-major arrays wrapped by the `mtrx` structure, with  $n_y$  rows of  $n_x$  values: element  $(i, j)$  lies at  $y = i \Delta y$ ,  $x = j \Delta x$  and has flat index  $i n_x + j$ , which is the ordering the Kronecker operators of eq. (17) assume. Earlier versions stored  $n_x$  rows of  $n_y$  values while the operators assumed the transpose; the two only agree when  $n_x = n_y$ , so non-square grids were rejected. Flattening a field is therefore a direct memory copy, and the two-dimensional derivative operations are implemented as sparse matrix–vector products.

#### 3.2 Grid convention

The grid is nodal. Column  $j$  lies at  $x_j = j \Delta x$ ,  $j = 0, \dots, n_x - 1$ , and row  $i$  at  $y_i = i \Delta y$ ,  $i = 0, \dots, n_y - 1$ , so the first and last rows and columns of the arrays lie on the walls and  $n_x$  nodes span  $L_x$  with  $n_x - 1$  intervals, as in eq. (9). All parts of the code use this convention:

1. wall velocities are imposed on the wall nodes, and the one-sided finite-difference closures are evaluated there;
2. all three Poisson solvers treat the  $(n_x - 2) \times (n_y - 2)$  interior nodes as the unknowns and set  $\psi = 0$  on the wall nodes (section 8.4);
3. centerline CSV output uses the node coordinates  $i \Delta y$  and  $j \Delta x$ , and interpolates linearly onto  $x = L_x/2$  (or  $y = L_y/2$ ) when no node lies on the centerline, i.e. for an even number of nodes.

Earlier versions mixed conventions: the spacing was  $L/n$ , the FFT solver treated all  $n_x \times n_y$  nodes as unknowns (placing  $\psi = 0$  one cell outside the walls, while Gauss–Seidel and SOR placed it on them), and centerline coordinates were labelled  $(i + \frac{1}{2})\Delta x$ . Switching between solvers changed the flow, and the centerline profiles of the default case differed from Ghia et al. by up to 0.06; see section 14.10.

### 4 Finite-Difference Spatial Discretization

#### 4.1 First derivative

The code supplies nominal second-, fourth-, and sixth-order centered stencils in the interior. For a uniform mesh and a sufficiently interior point,

### Second order

$$f'_i \approx \frac{f_{i+1} - f_{i-1}}{2\Delta x} + \mathcal{O}(\Delta x^2). \quad (10)$$

### Fourth order

$$f'_i \approx \frac{f_{i-2} - 8f_{i-1} + 8f_{i+1} - f_{i+2}}{12\Delta x} + \mathcal{O}(\Delta x^4). \quad (11)$$

### Sixth order

$$f'_i \approx \frac{-f_{i-3} + 9f_{i-2} - 45f_{i-1} + 45f_{i+1} - 9f_{i+2} + f_{i+3}}{60\Delta x} + \mathcal{O}(\Delta x^6). \quad (12)$$

These coefficients correspond directly to the entries created by `SDiff1()`.

## 4.2 Second derivative

For the Laplacian terms in the vorticity transport equation, `SDiff2()` implements:

### Second order

$$f''_i \approx \frac{f_{i-1} - 2f_i + f_{i+1}}{\Delta x^2} + \mathcal{O}(\Delta x^2). \quad (13)$$

### Fourth order

$$f''_i \approx \frac{-f_{i-2} + 16f_{i-1} - 30f_i + 16f_{i+1} - f_{i+2}}{12\Delta x^2} + \mathcal{O}(\Delta x^4). \quad (14)$$

### Sixth order

$$f''_i \approx \frac{2f_{i-3} - 27f_{i-2} + 270f_{i-1} - 490f_i + 270f_{i+1} - 27f_{i+2} + 2f_{i+3}}{180\Delta x^2} + \mathcal{O}(\Delta x^6). \quad (15)$$

## 4.3 Boundary closure and formal order

The requested `order` controls the widest interior stencil, but the boundary rows deliberately use narrower formulas. The closures are summarized in table 1.

Table 1: Effective stencil order near an array boundary in the current finite-difference matrices. The pattern is mirrored at the opposite boundary.

| Requested order | Location            | First derivative   | Second derivative   |
|-----------------|---------------------|--------------------|---------------------|
| 2               | boundary row        | 1st-order forward  | 2nd-order one-sided |
| 2               | interior            | 2nd-order centered | 2nd-order centered  |
| 4               | boundary row        | 1st-order forward  | 2nd-order one-sided |
| 4               | first interior row  | 2nd-order centered | 2nd-order centered  |
| 4               | deeper interior     | 4th-order centered | 4th-order centered  |
| 6               | boundary row        | 1st-order forward  | 2nd-order one-sided |
| 6               | first interior row  | 2nd-order centered | 2nd-order centered  |
| 6               | second interior row | 4th-order centered | 4th-order centered  |
| 6               | deeper interior     | 6th-order centered | 6th-order centered  |

Therefore, statements such as “sixth-order spatial discretization” should be read as *sixth-order centered interior stencils*, not as a demonstrated sixth-order global convergence rate. The global rate depends on the boundary closure, the Poisson discretization, the solution regularity, and the grid convention discussed above. Measured against a smooth manufactured solution with stationary no-slip walls, it is second order for every derivative order (section 14.5).

## 4.4 Two-dimensional operators by Kronecker product

Let  $d_x, d_y$  denote the one-dimensional first-derivative matrices,  $d_{xx}$  and  $d_{yy}$  the one-dimensional second-derivative matrices, and  $I_x, I_y$  identity matrices. The code assembles

$$D_x = I_y \otimes d_x, \quad D_y = d_y \otimes I_x, \quad (16)$$

$$D_{xx} = I_y \otimes d_{xx}, \quad D_{yy} = d_{yy} \otimes I_x. \quad (17)$$

This is a convenient algebraic representation of tensor-product finite differences on a Cartesian grid.

The one-dimensional matrices are assembled directly in compressed sparse row (CSR) form, and so are the two-dimensional Kronecker products. For the widest sixth-order stencil, each derivative row has only a small fixed number of nonzero entries, so sparse matrix–vector multiplication scales linearly with the number of grid points rather than quadratically or cubically in the matrix dimension.

## 5 Boundary Conditions and Wall Vorticity

### 5.1 Velocity boundary conditions

The `wall\_bc` structure stores a constant  $(u, v)$  value for each of four array edges. At the beginning of every timestep, `apply\_wall\_bc()` writes those values directly into the edge rows and columns of the velocity arrays.

For the default cavity,

$$\mathbf{u}_{\text{lid}} = (1, 0), \quad \mathbf{u}_{\text{other walls}} = (0, 0). \quad (18)$$

No-slip and no-penetration are therefore represented through prescribed wall velocity.

### 5.2 Vorticity boundary treatment

A vorticity–streamfunction method requires a boundary value for  $\omega$  that is consistent with the velocity boundary condition. The current code computes the velocity derivatives with the same finite-difference operators used elsewhere,

$$\omega|_{\partial\Omega} = (D_x v - D_y u)|_{\partial\Omega}, \quad (19)$$

and overwrites the boundary entries of  $\omega$  before advancing the field, again at every RK4 stage, and once more at the end of the step. The update itself advances the boundary entries with the transport equation, which does not hold there, so the last overwrite replaces them with the wall vorticity of the new velocity; the field a step returns, and the one written to the VTK files, is consistent with its velocity at the walls too.

This differs from the frequently used Thom-type wall formula, which derives wall vorticity directly from a neighboring streamfunction value and wall velocity. In `cnavier`, the wall vorticity is instead obtained through one-sided/narrow derivative rows acting on the explicitly imposed velocity field. This design has the advantage of reusing one operator family for both interior and boundary derivatives, but it also means that the low-order boundary closure in table 1 directly influences the wall vorticity accuracy.

With `-wall-closure briley (solver_config.wall_closure = 1)`, the wall vorticity is instead taken from the stream function with Briley’s third-order formula. Along a wall  $\psi = 0$ , so  $\omega = -\partial^2 \psi / \partial s^2$  there,  $s$  the distance from the wall; from  $\psi_0, \dots, \psi_3$  at  $s = 0, h, 2h, 3h$  and the wall-tangential velocity  $U = \partial \psi / \partial s(0)$ ,

$$\omega_0 = \frac{85\psi_0 - 108\psi_1 + 27\psi_2 - 4\psi_3}{18h^2} + \frac{11U}{3h}, \quad (20)$$

with  $U = u$  on the bottom wall,  $-u$  on the top,  $-v$  on the left and  $v$  on the right. Thom's formula,  $\omega_0 = 2(\psi_0 - \psi_1)/h^2 + 2U/h$ , is the first-order member of the same family. It is used at every RK4 stage and at the end of a step like the velocity-based formula; a first step derives  $\psi$  from  $\omega$  before using it. Section 14.7 measures what it changes.

### 5.3 Streamfunction boundary condition

The Poisson solvers impose homogeneous Dirichlet conditions on the streamfunction. For a closed cavity, setting a constant value of  $\psi$  on every wall is consistent with no normal flow; the arbitrary constant is chosen to be zero:

$$\psi|_{\partial\Omega} = 0. \quad (21)$$

Tangential wall motion is then communicated to the interior through the vorticity boundary values rather than through a nonzero streamfunction value.

## 6 Vorticity Transport Discretization

Define the semi-discrete right-hand side

$$\mathcal{F}(\omega; u, v) = -u \odot (D_x \omega) - v \odot (D_y \omega) + \frac{1}{\text{Re}} (D_{xx} \omega + D_{yy} \omega), \quad (22)$$

where  $\odot$  denotes pointwise multiplication. This is exactly the operation implemented in the CPU `dwdt()` routine and in the CUDA `rhs\_kernel`.

The convective terms are centered in the interior for all three selectable orders. Centered advection is nondissipative, which is attractive for accuracy but provides no built-in upwind damping. Consequently, timestep choice and adequate viscous resolution become important, particularly as  $\text{Re}$  increases.

## 7 Time Integration

### 7.1 Forward Euler

With `time\_scheme = 1`, the update is

$$\omega^{n+1} = \omega^n + \Delta t \mathcal{F}(\omega^n; u^n, v^n). \quad (23)$$

After the explicit vorticity update, the Poisson equation is solved once and the velocity is reconstructed from the new streamfunction. Thus Euler requires one elliptic solve per full timestep.

Forward Euler is first-order accurate in time. Its stability region does not include a finite segment of the imaginary axis, so centered advection by itself is not Euler-stable. Viscosity may stabilize a particular discretization and timestep, but the method should be treated as the low-cost/low-order option rather than the recommended high-Reynolds-number integrator.

## 7.2 Classical fourth-order Runge–Kutta

With `time\_scheme = 2`, the code uses classical RK4:

$$k_1 = \mathcal{F}(\omega^n), \quad (24)$$

$$k_2 = \mathcal{F}\left(\omega^n + \frac{\Delta t}{2}k_1\right), \quad (25)$$

$$k_3 = \mathcal{F}\left(\omega^n + \frac{\Delta t}{2}k_2\right), \quad (26)$$

$$k_4 = \mathcal{F}(\omega^n + \Delta t k_3), \quad (27)$$

$$\omega^{n+1} = \omega^n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (28)$$

The velocity depends on the stage vorticity through eq. (6)–eq. (5). Accordingly, every call to the RK4 right-hand side performs a Poisson solve and recovers a stage-consistent  $(u, v)$  field. Four Poisson solves are therefore required for the four RK stages, followed by one final solve to make  $(u^{n+1}, v^{n+1})$  consistent with  $\omega^{n+1}$ . In total, RK4 performs five Poisson solves per timestep.

For a sufficiently smooth semi-discrete system, classical RK4 is fourth-order accurate in time. In the complete method that holds only if the boundary data move with the stages: the wall vorticity eq. (19) depends on the velocity, which changes from stage to stage, so it is recomputed from each stage’s velocity before that stage’s derivatives are taken (section 5.2). Earlier versions set it once per timestep, from the velocity at the start of the step. The boundary data then lagged by up to one step, RK4 converged at first order only, and at the same step its error was about that of Euler. Measured on the lid-driven cavity ( $17 \times 17$ ,  $\text{Re} = 100$ ,  $\omega$  on all nodes, walls included, at  $t = 0.2$  against a run of the same scheme at  $\Delta t/64$ ), the observed order is now 4.0 for RK4 and 1.0 for Euler, and at  $\Delta t = T/64$  the error of RK4 is  $1.2 \times 10^{-8}$  against  $2.5 \times 10^{-2}$  for Euler. The test suite checks both orders. The steady state is essentially unaffected: at  $t = 30$  the centerline profiles of the default case move by at most  $1.2 \times 10^{-4}$ , and their distance from the Ghia data changes from 0.0024/0.0074 to 0.0023/0.0073 ( $u/v$ ).

## 8 Poisson Equation for the Streamfunction

At each velocity reconstruction, the code solves

$$\nabla^2 \psi = -\omega. \quad (29)$$

Three algorithms are available.

### 8.1 Five-point discrete Poisson operator

The iterative solvers and the spectral eigenvalues correspond to the conventional second-order five-point Laplacian. At an interior point, with  $i$  the row ( $y$ ) and  $j$  the column ( $x$ ) index as everywhere in this document,

$$\frac{\psi_{i,j+1} - 2\psi_{i,j} + \psi_{i,j-1}}{\Delta x^2} + \frac{\psi_{i+1,j} - 2\psi_{i,j} + \psi_{i-1,j}}{\Delta y^2} = f_{i,j}, \quad f = -\omega. \quad (30)$$

This is an important distinction: selecting fourth- or sixth-order finite-difference operators for vorticity transport and velocity differentiation does *not* change the order of the Poisson operator. The streamfunction solve remains based on the second-order discrete Laplacian.

## 8.2 Gauss–Seidel

Solving eq. (30) for the center value gives the Gauss–Seidel update

$$\psi_{i,j}^{\text{GS}} = \frac{\Delta y^2(\psi_{i,j+1} + \psi_{i,j-1}) + \Delta x^2(\psi_{i+1,j} + \psi_{i-1,j}) - \Delta x^2 \Delta y^2 f_{i,j}}{2(\Delta x^2 + \Delta y^2)}. \quad (31)$$

The serial CPU build sweeps lexicographically. The OpenMP build uses red–black ordering so that all nodes of one color can be updated in parallel. The CUDA implementation also uses red–black ordering.

## 8.3 Successive over-relaxation

SOR applies a relaxation factor  $\beta$ :

$$\psi_{i,j}^{(k+1)} = \beta \psi_{i,j}^{\text{GS}} + (1 - \beta) \psi_{i,j}^{(k)}. \quad (32)$$

The default program estimates an optimal relaxation parameter from the spectral radius of Jacobi iteration on the  $(n_x - 2) \times (n_y - 2)$  interior nodes. With the weights of the five-point stencil ( $\Delta y^2$  on the  $x$  neighbours,  $\Delta x^2$  on the  $y$  neighbours), the slowest Jacobi mode is the lowest sine mode in each direction, so

$$\rho = \frac{\Delta y^2 \cos\left(\frac{\pi}{n_x - 1}\right) + \Delta x^2 \cos\left(\frac{\pi}{n_y - 1}\right)}{\Delta x^2 + \Delta y^2}, \quad (33)$$

$$\beta = \frac{2}{1 + \sqrt{1 - \rho^2}}. \quad (34)$$

which reduces to  $\rho = \cos(\pi/(n - 1))$  on a square grid with  $\Delta x = \Delta y$ . The five-point stencil is consistently ordered in both the lexicographic and the red–black sweep, so Young’s optimum  $\beta$  applies to the CPU and the GPU solvers alike.

The default maximum iteration count is 10000 and the shipped convergence threshold is  $10^{-3}$ .

**Stopping metric.** The implementation function named `error()` computes

$$E^{(k)} = \sum_{i,j} \left| \psi_{i,j}^{(k+1)} - \psi_{i,j}^{(k)} \right|, \quad (35)$$

not a residual norm and not a root-sum-square norm. The console string currently labels this quantity “RSS error”, but scientifically it is an  $L_1$  iterate-change metric. Verification tests separately compute the actual five-point Poisson residual.

Because the default tolerance can stop the iteration before full convergence, the sweep ordering can affect the returned approximation. This is why the serial lexicographic CPU result and the red–black GPU result need not be identical at the shipped tolerance; they approach the same solution as the tolerance is tightened.

## 8.4 Direct FFT/DST-I solver

The default `poisson\_type = 3` uses a two-dimensional discrete sine transform of type I (DST-I) of the  $m_x \times m_y$  interior nodes,  $m_x = n_x - 2$ ,  $m_y = n_y - 2$ . A DST-I of length  $m$  implies zeros at indices  $-1$  and  $m$ , which are exactly the wall nodes, so the solver imposes  $\psi = 0$  on the wall nodes like the iterative solvers and solves the same discrete problem. For each spectral mode  $(p, q)$ , the standard second-difference Laplacian has eigenvalue

$$\lambda_{p,q} = \frac{2 \cos\left(\frac{\pi p}{m_x + 1}\right) - 2}{\Delta x^2} + \frac{2 \cos\left(\frac{\pi q}{m_y + 1}\right) - 2}{\Delta y^2}, \quad (36)$$

with  $p = 1, \dots, m_x$  and  $q = 1, \dots, m_y$ ; note  $m_x + 1 = n_x - 1$ .

The algorithm is:

1. transform the right-hand side with a two-dimensional DST-I;
2. divide each transformed coefficient by  $\lambda_{p,q}$ ;
3. apply the DST-I again;
4. multiply by the inverse transform normalization

$$\frac{1}{4(m_x + 1)(m_y + 1)} = \frac{1}{4(n_x - 1)(n_y - 1)}.$$

On the CPU the two-dimensional DST-I is applied as two passes of one-dimensional FFTW3 R0DFT00 transforms, first along the rows and then along the columns, in batches of eight rows or columns. The plans are created once at startup with `FFTW\_ESTIMATE` and reused. Two choices make the result deterministic. `FFTW\_ESTIMATE` picks plans without timing them; `FFTW\_MEASURE`, used previously, could pick a different plan on every run, and different plans round differently, so repeated runs at  $513 \times 513$  differed in the last bits. And every row is transformed by the same plan whatever the number of threads, so serial and OpenMP builds give bit-identical results. The asymptotic cost is  $\mathcal{O}(N \log N)$  with  $N = n_x n_y$ , compared with a potentially large number of  $\mathcal{O}(N)$  sweeps for Gauss–Seidel or SOR.

Once Gauss–Seidel or SOR has converged, all three solvers return the same  $\psi$ ; the test suite checks agreement to  $10^{-9}$  on a random right-hand side.

The FFT solver is “direct” with respect to the *discrete second-order Poisson problem*: once the transform is performed, no iterative convergence tolerance is required. It is not an exact solver for the continuous Poisson equation; the continuous truncation error remains that of the discrete Laplacian and the grid representation.

## 8.5 Fourth-order compact Poisson operator

With `-poisson-order 4 (solver_config.poisson_order)`, the FFT solver discretises  $\nabla^2 \psi = f$  with the compact nine-point (“Mehrstellen”) operator instead of the five-point one,

$$\left[ \delta_x^2 + \delta_y^2 + \frac{\Delta x^2 + \Delta y^2}{12} \delta_x^2 \delta_y^2 \right] \psi = \left[ 1 + \frac{\Delta x^2 \delta_x^2 + \Delta y^2 \delta_y^2}{12} \right] f, \quad (37)$$

with  $\delta_x^2, \delta_y^2$  the second differences of eq. (30). It is fourth order for any  $\Delta x, \Delta y$ . The same sine transform diagonalises it: mode  $(p, q)$  has eigenvalue  $\mu_p + \mu_q + \frac{\Delta x^2 + \Delta y^2}{12} \mu_p \mu_q$ , where  $\mu_p$  and  $\mu_q$  are the two terms of eq. (36), the eigenvalues of  $\delta_x^2$  and  $\delta_y^2$ , so the cost is that of the five-point solve. The right-hand side is  $f + (f_E + f_W + f_N + f_S - 4f)/12$ , whatever the spacing. Next to a wall it needs  $f = -\omega$  on the wall nodes; these are taken from the quadratic through the first three interior nodes along the wall normal,  $3f_1 - 3f_2 + f_3 = f_0 + \mathcal{O}(h^3)$ , rather than from the wall vorticity, so the Poisson solve still reads only the interior of  $\omega$  and does not couple back to the wall closure. The extrapolation error,  $\mathcal{O}(h^3)$  in the first interior row only, does not lower the order. On a test with a non-zero Laplacian on the walls the operator converges at about five (4.9 from  $33^2$  to  $129^2$ ). The option needs the FFT solver; Gauss–Seidel and SOR keep the five-point operator.

## 8.6 Doubly periodic boundaries

With `solver_config.periodic` set, the domain  $[0, L_x) \times [0, L_y)$  is periodic in both directions and has no walls. The grid has  $n_x$  points spaced  $\Delta x = L_x/n_x$  (the point at  $x = L_x$  is the point at  $x = 0$ ), and likewise in  $y$ . Three things change.

- **Derivative operators.** Every row of `SDiff1_periodic` and `SDiff2_periodic` is the centered interior stencil of the chosen order, with the indices taken modulo  $n$ , so there are no one-sided rows.
- **Poisson operator.** The Laplacian is  $D_{xx} + D_{yy}$ , built from the same second-derivative operators as the transport, so the Poisson solve has the order of the derivatives rather than second order. Both operators are circulant, so the two-dimensional discrete Fourier transform diagonalises them: mode  $(k_x, k_y)$  has eigenvalue  $\lambda_x(k_x) + \lambda_y(k_y)$ , where  $\lambda_x(k) = \sum_m c_m \cos(2\pi km/n_x)$  is the symbol of the stencil with coefficients  $c_m$ , read from the operator's first row. The solver transforms  $-\omega$  with a real-to-complex FFT (FFTW, or cuFFT on the GPU), divides by the eigenvalues, and transforms back. The mean mode has eigenvalue zero: the mean of  $\psi$  is set to zero and the mean of  $\omega$  is ignored, as a periodic domain carries no net circulation.
- **No wall terms.** The wall-velocity and wall-vorticity updates are skipped. The periodic grids need the FFT solver; Gauss-Seidel and SOR are not offered.

The command line provides two periodic cases on the unit square: the Taylor–Green vortex  $\psi = \sin(kx) \sin(ky) e^{-2k^2 t/\text{Re}}/k$ ,  $k = 2\pi$ , an exact solution whose error the program prints at the end, and the double shear layer of Bell, Colella and Glaz (1989), which rolls up into vortices at high Reynolds numbers.

## 9 Complete Timestep Algorithms

### 9.1 Euler sequence

One CPU Euler timestep performs the following operations:

1. impose wall velocities on  $u$  and  $v$ ;
2. compute  $D_y u$  and  $D_x v$ ;
3. overwrite boundary vorticity using  $\omega = D_x v - D_y u$ ;
4. compute  $D_x \omega$ ,  $D_y \omega$ ,  $D_{xx} \omega$ , and  $D_{yy} \omega$ ;
5. update  $\omega$  by forward Euler;
6. solve  $\nabla^2 \psi = -\omega$ ;
7. recover  $u = D_y \psi$  and  $v = -D_x \psi$ ;
8. impose the wall velocities on copies of  $u$  and  $v$  and overwrite the boundary vorticity with  $D_x v - D_y u$  of those copies, since the update advanced the boundary entries with the transport equation, which does not hold there.  $u$  and  $v$  themselves keep the values from  $\psi$ , whose divergence the continuity check measures.

### 9.2 RK4 sequence

One RK4 timestep begins with the same wall-velocity and wall-vorticity update. Each of the four stage right-hand-side evaluations then:

1. solves the Poisson equation for the stage streamfunction from the interior stage vorticity;
2. recovers a stage-consistent velocity field and imposes the wall velocities on it;
3. sets the wall vorticity of the stage from that velocity, eq. (19);

4. differentiates the stage vorticity;
5. evaluates the nonlinear convective and viscous terms.

After combining  $k_1$  through  $k_4$ , a fifth Poisson solve recovers velocity for the completed step, and the step ends like the Euler step: the wall velocities are imposed on copies of  $u$  and  $v$ , and the boundary vorticity is overwritten with  $D_x v - D_y u$  of those copies.

### 9.3 Pseudocode

```

construct sparse DX, DY, DX2, DY2
allocate reusable solver workspace
initialize u, v, omega

for each timestep:
    impose wall velocities on u, v
    omega_wall = DX*v - DY*u on array edges

    if Euler:
        rhs = -u*(DX*omega) - v*(DY*omega)
              + (1/Re)*(DX2*omega + DY2*omega)
        omega += dt*rhs
        solve Laplacian(psi) = -omega
        u = DY*psi
        v = -DX*psi

    if RK4:
        for each RK stage:
            solve Laplacian(psi_stage) = -omega_stage    (interior)
            u_stage = DY*psi_stage
            v_stage = -DX*psi_stage
            impose wall velocities on u_stage, v_stage
            omega_stage_wall = DX*v_stage - DY*u_stage on array edges
            k_stage = RHS(omega_stage, u_stage, v_stage)
        omega = RK4 combination
        solve Laplacian(psi) = -omega
        u = DY*psi
        v = -DX*psi

    (u_w, v_w) = (u, v) with wall velocities imposed
    omega_wall = DX*v_w - DY*u_w on array edges

    evaluate discrete divergence DX*u + DY*v
    optionally write VTK

```

## 10 Stability and Timestep Selection

Because both time integrators are explicit, the timestep is restricted by the spectra of the discrete advection and diffusion operators.

### 10.1 Implemented Courant check

Before starting a run, `main.c` computes, for the default moving-wall velocity,

$$C_x = U_{\text{lid}} \frac{\Delta t}{\Delta x}, \quad C_y = U_{\text{lid}} \frac{\Delta t}{\Delta y}, \quad (38)$$

and aborts if either exceeds `max\_co = 1`, where  $U_{\text{lid}}$  is the largest wall speed. This is a useful sanity check but should not be interpreted as a complete stability proof. The actual interior velocity can differ from the lid velocity, and diffusion imposes a separate explicit restriction, which is checked as described next.

## 10.2 Implemented viscous limit

`max\_stable\_dt()` in `fluiddyn.c` bounds the timestep for the viscous term  $\text{Re}^{-1}(D_{xx} + D_{yy})$ . The interior second-derivative stencils are centered with coefficients of alternating sign, so the sum of their magnitudes is the value of the stencil symbol at the highest grid frequency, i.e. the largest eigenvalue magnitude:

$$\lambda_{\max} = \frac{1}{\text{Re}} \left( \sum_k |a_k^x| + \sum_k |a_k^y| \right), \quad (39)$$

taken from the middle row of the one-dimensional operators (which is also the middle row of the two-dimensional ones), so the check runs before the two-dimensional operators are assembled. The run aborts unless

$$\Delta t \leq \frac{\rho}{\lambda_{\max}}, \quad \rho = \begin{cases} 2 & \text{forward Euler,} \\ 2.785293\dots & \text{classical RK4,} \end{cases} \quad (40)$$

$\rho$  being the extent of each method's stability region along the negative real axis, and unless the Courant condition holds; the message names both limits and suggests the smaller one, rounded down so that the suggestion itself passes. For forward Euler the classical centered advection–diffusion condition  $\Delta t \leq 2\nu/U^2 = 2/(\text{Re } U^2)$  is checked as well, but only as a warning, since it is conservative here (below); when an Euler run is refused, the suggested  $\Delta t$  is the smallest of all three limits, so that following it does not trigger the warning. Measured on the lid-driven cavity, the viscous bound is sharp: for orders 2, 4 and 6, both schemes and  $n = 64$  and 128, runs at 0.95 times the bound stay bounded for 3000 steps and runs at 1.05 times diverge. The advection condition is conservative: at  $\text{Re} = 1000$ ,  $n = 64$ , Euler stayed bounded up to about three times it. At the defaults ( $n = 64$ ,  $\text{Re} = 100$ , sixth order) the bounds are  $\Delta t \leq 0.00580$  for RK4 and  $\Delta t \leq 0.00416$  for Euler.

## 10.3 Diffusive scaling

For the simple second-order Laplacian with forward Euler, the familiar two-dimensional diffusion restriction is approximately

$$\frac{1}{\text{Re}} \Delta t \left( \frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right) \lesssim \frac{1}{2}. \quad (41)$$

For  $\Delta x = \Delta y = h$ , this reduces to

$$\Delta t \lesssim \frac{\text{Re } h^2}{4}. \quad (42)$$

The precise limit changes with the higher-order second-derivative stencil and with RK4, but the important scaling remains

$$\Delta t = \mathcal{O}(h^2) \quad (43)$$

when explicit viscous stability controls the calculation. The repository benchmarks therefore use a timestep proportional to  $1/n^2$ .

## 10.4 Centered advection

The interior convective derivative is centered. Such a discretization contributes nearly imaginary eigenvalues for pure linear advection. RK4 has a finite stability interval along the imaginary axis and is therefore much better suited than Euler to centered advection. At high Reynolds number, however, neither time integrator substitutes for spatial stabilization if the grid is too coarse to resolve the vorticity gradients.

## 10.5 Recommended practice

For scientific runs:

- prefer RK4 over Euler unless Euler is being used deliberately for a comparison;
- reduce  $\Delta t$  under grid refinement, normally at least proportionally to  $h^2$  until temporal error is demonstrated to be negligible;
- verify timestep independence by repeating selected calculations with  $\Delta t/2$ ;
- monitor divergence, extrema, and visual smoothness of the vorticity field;
- treat a run that merely passes the implemented  $C \leq 1$  check as *admissible input*, not automatically as a stable or converged simulation.

## 11 Sparse Linear Algebra

The dominant derivative operation is a CSR sparse matrix–vector multiplication

$$\mathbf{y} = A\mathbf{x}. \quad (44)$$

For each row, `spmv()` loops only over the nonzero entries stored between consecutive elements of `row_ptr`. Since the finite-difference bandwidth is fixed, the cost is  $\mathcal{O}(N)$  for  $N = n_x n_y$ .

The main dense field arrays are contiguous. Reusable flat buffers and RK4 work arrays are allocated once before the time loop; no per-stage dynamic memory allocation is needed. This improves both CPU cache behavior and GPU transfer strategy.

## 12 Parallel Implementations

### 12.1 Backend structure

A run is described by one `solver_config` (`fluiddyn.h`): grid size and spacing, Reynolds number, timestep, time scheme, Poisson solver settings, wall velocities and the four derivative operators. Both backends are initialised from it, and each keeps its own copy, taken at that moment: the CPU solver in its workspace (`rk4_ctx`, which also holds the work arrays and FFT plans) and the GPU solver (`gpu_init`), which uploads the operators and fields to the device. Changing the caller’s `solver_config` afterwards affects neither, and a test checks that both ignore such a change alike. `backend.c` chooses between them once, at startup, and exposes the same operations for both (one timestep, the divergence diagnostic, the host fields), so the time loop in `main.c` does not depend on where it runs. On the CPU the host fields are the solver’s own arrays; on the GPU they are copies filled from the device when asked for. A GPU run allocates no CPU solver workspace, and the memory check counts only what the chosen backend allocates.

## 12.2 OpenMP

Building with

```
make OPENMP=1
```

enables shared-memory parallel loops in the principal CSR operations, explicit right-hand-side loops, and suitable Poisson operations, including the batches of one-dimensional sine transforms. Loops over fewer than 2048 grid points (`OMP\_MIN\_WORK`) stay serial, since starting the threads would cost more than they save. Unless `OMP\_NUM\_THREADS` or a thread placement (`OMP\_PROC\_BIND`, `OMP\_PLACES`) is set, the program uses one thread per physical core in its affinity mask instead of the OpenMP default of one per logical CPU: with two hardware threads per core the second adds nothing to these bandwidth-limited loops, and on an idle machine 16 threads on 8 cores were 25–40% slower than 8. Surplus threads are far more costly when other programs occupy the cores, since every parallel loop ends at a barrier where all threads wait for the slowest. Because the CSR products and transforms are limited by memory bandwidth, the speed-up levels off at about two from four threads on. Iterative elliptic solves use red–black coloring because a naive parallelization of lexicographic Gauss–Seidel would violate its data dependence.

Changing the sweep order is numerically relevant when the solver is stopped before convergence. Therefore, at a loose iterate-change tolerance, serial Gauss–Seidel/SOR and OpenMP red–black Gauss–Seidel/SOR can return slightly different intermediate approximations even though they converge to the same discrete solution when the tolerance is sufficiently tight.

## 12.3 CUDA

Building with

```
make CUDA=1
```

adds a GPU backend. The four CSR derivative matrices and all primary fields are uploaded once and remain on the device throughout the time loop. CUDA kernels implement:

- wall velocity and wall-vorticity updates;
- CSR sparse derivatives;
- Euler and RK4 stage algebra;
- velocity recovery;
- red–black Gauss–Seidel/SOR;
- the FFT-based Poisson solver;
- the divergence diagnostic and reductions.

All solver arithmetic is double precision.

## 12.4 GPU sine transform through odd extension

cuFFT does not expose a native DST-I. The CUDA backend uses the identity that the DST-I of  $x_1, \dots, x_m$  is  $-\text{Im}$  of the real FFT of the odd extension  $(0, x_1, \dots, x_m, 0, -x_m, \dots, -x_1)$  of length  $2(m+1)$ . The two-dimensional transform is done in two passes of batched one-dimensional transforms, as on the CPU: the odd extensions of the  $n_y - 2$  interior rows (length  $2(n_x - 1)$ ) are transformed together, the sine modes are read off and extended along the columns (length  $2(n_y - 1)$ ), and those are transformed together. The code then divides by the same eigenvalues

as eq. (36), applies the transform again, and scales by the DST normalization. An earlier version transformed one two-dimensional odd extension of size  $2(n_y - 1) \times 2(n_x - 1)$ , four times the grid; the batched passes are about 1.3 times faster from  $129 \times 129$  upwards.

The red-black Gauss-Seidel and SOR solvers evaluate their stopping rule on the device: each sweep's sum of  $|\Delta\psi|$  is reduced and compared with the tolerance by a kernel, sweeps after the one that meets it do nothing, and the host reads the outcome once every 32 sweeps. The iteration count and the result are the same as with a check after every sweep.

This makes the GPU FFT algorithm mathematically equivalent to the CPU FFTW DST-I solver, up to floating-point round-off and the details of the FFT implementations.

## 13 Diagnostics and Output

### 13.1 Discrete continuity diagnostic

After every timestep, the CPU path computes

$$\epsilon \nabla \cdot u = D_x u + D_y v \quad (45)$$

and prints its maximum and minimum. The CUDA path computes the same field on-device and reduces it to its extrema. Since velocity is recovered from a streamfunction, these values are expected to remain close to floating-point round-off for the compatible operator pair.

This diagnostic should be interpreted as a consistency check on the streamfunction-to-velocity machinery. It is not, by itself, evidence that the momentum/vorticity equations are spatially converged.

### 13.2 VTK output

By default, vorticity is written to legacy ASCII VTK files in `output/`. The output interval is configurable and may be disabled with

```
./cnavier --output-interval 0
```

for performance measurements.

### 13.3 Centerline profiles

At the end of a run, the code writes four CSV files:

- `centerline\_u\_sim.csv`;
- `centerline\_u\_ghia.csv`;
- `centerline\_v\_sim.csv`;
- `centerline\_v\_ghia.csv`.

The reference data are the classical lid-driven-cavity values of Ghia, Ghia, and Shin at  $Re = 100$  [1]. The project exports both numerical and reference profiles, but does not currently compute an automated interpolation-based error norm between them.

### 13.4 Flow diagnostics

After every step the program appends to `output/integrals.csv` the domain means of the kinetic energy, the enstrophy and the palinstrophy,

$$E = \frac{1}{2} \langle u^2 + v^2 \rangle, \quad Z = \frac{1}{2} \langle \omega^2 \rangle, \quad P = \frac{1}{2} \langle |\nabla \omega|^2 \rangle, \quad (46)$$

with  $\nabla\omega$  from  $D_x, D_y$ . On a periodic grid  $\langle \cdot \rangle$  is the mean over the nodes; with walls it is the trapezoidal mean over the domain, with the wall velocities on the wall nodes. The GPU computes them by reductions on the device, so no field is copied to the host for them. In an unforced periodic flow the continuum budgets are  $dE/dt = -2\nu Z$  and  $dZ/dt = -2\nu P$ .

On a periodic grid, each VTK frame is accompanied by `output/spectrum-1-<n>.csv` with the shell-averaged spectra and spectral fluxes. Mode  $\mathbf{k} = (2\pi m/L_x, 2\pi n/L_y)$  falls in shell  $b$  when  $|b\Delta k - |\mathbf{k}|| \leq \Delta k/2$ ,  $\Delta k = 2\pi/\max(L_x, L_y)$ , and

$$E(k_b) = \frac{1}{2} \sum_{\mathbf{k} \in b} (|\hat{u}_{\mathbf{k}}|^2 + |\hat{v}_{\mathbf{k}}|^2), \quad Z(k_b) = \frac{1}{2} \sum_{\mathbf{k} \in b} |\hat{\omega}_{\mathbf{k}}|^2, \quad (47)$$

with the transforms normalised so that  $\sum_b E(k_b) = E$  and  $\sum_b Z(k_b) = Z$ . The fluxes through  $k_b$  are

$$\Pi_E(k_b) = - \sum_{b' \leq b} \sum_{\mathbf{k} \in b'} \text{Re}(\hat{\psi}_{\mathbf{k}}^* \hat{N}_{\mathbf{k}}), \quad \Pi_Z(k_b) = - \sum_{b' \leq b} \sum_{\mathbf{k} \in b'} \text{Re}(\hat{\omega}_{\mathbf{k}}^* \hat{N}_{\mathbf{k}}), \quad (48)$$

where  $N = -(uD_x\omega + vD_y\omega)$  is the solver's own nonlinear term and  $\hat{\psi}$  solves  $(D_{xx} + D_{yy})\psi = -\omega$  with the solver's eigenvalues (section 8.6), so the fluxes are those of the discrete equations. A positive flux carries energy or enstrophy towards larger  $k$ . In the continuum the nonlinear term conserves both, so  $\Pi_E$  and  $\Pi_Z$  vanish beyond the largest shell; in the discrete equations they do so up to the order of the derivatives (section 14.9).

## 14 Verification and Validation

It is useful to separate *verification* from *validation*. Verification asks whether the implementation solves its stated discrete equations correctly. Validation asks whether those equations reproduce accepted physical/reference results to the required accuracy.

### 14.1 Analytic verification of the FFT Poisson solver

The test suite constructs a discrete sine eigenmode

$$f_{ij} = \sin\left(\frac{\pi p i}{n-1}\right) \sin\left(\frac{\pi q j}{n-1}\right) \quad (49)$$

on the nodes  $i, j = 0, \dots, n-1$  (it vanishes on the wall nodes), with  $n = 32$ ,  $p = 3$ , and  $q = 5$ . Since this is an exact eigenvector of the discrete Laplacian, the exact discrete solution is

$$\psi = \frac{f}{\lambda_{p,q}}. \quad (50)$$

The test requires the maximum relative difference to be no larger than  $10^{-10}$ . This directly verifies the transform, eigenvalue division, and normalization.

### 14.2 Residual verification of the iterative Poisson solvers

For random right-hand-side data on a  $24 \times 24$  grid, Gauss-Seidel and SOR are run with a tight iterate-change tolerance of  $10^{-9}$ . The test then evaluates the actual five-point residual

$$r_{ij} = (L_h\psi)_{ij} - f_{ij} \quad (51)$$

and requires

$$\|r\|_{\infty} \leq 10^{-4}. \quad (52)$$

This is valuable because the production stopping criterion is based on successive iterate change rather than directly on  $r$ .

### 14.3 Short cavity integration tests

Both Euler and RK4 are exercised for 50 timesteps on a  $32 \times 32$  lid-driven cavity at  $\text{Re} = 100$ , with sixth-order interior derivative operators, the FFT Poisson solver, and  $\Delta t = 0.002$ . The tests require:

- finite fields with bounded vorticity magnitude;
- a nontrivial velocity response to the moving lid;
- maximum discrete divergence no larger than  $10^{-9}$ .

These are integration/smoke tests rather than accuracy tests against a fully converged reference solution.

### 14.4 CPU–GPU equivalence tests

When built with CUDA and run on a usable GPU, the test suite checks:

- host/device field round trips;
- continuity reductions on grids up to  $300 \times 300$ ;
- all four CSR derivative operators on several grid sizes;
- FFT, SOR, and Gauss–Seidel Poisson solutions;
- Euler and RK4 full timesteps;
- multiple grid sizes, including  $128 \times 128$ ;
- a case with four independently moving walls.

Typical comparison thresholds in the tests are  $10^{-12}$  for CSR and FFT building blocks,  $10^{-11}$  for full FFT-based timesteps, and  $10^{-7}$  for converged iterative-solver comparisons. With OpenMP enabled, additional tests compare CPU and GPU red–black iterative solvers at the shipped production tolerance.

A CUDA test build with no usable GPU returns exit status 77 rather than reporting a false pass.

### 14.5 Spatial convergence against a manufactured solution

The cavity cannot measure the spatial order of the method: the lid is discontinuous at the top corners, the vorticity is singular there, and refinement on the cavity measures the singularity. The Ghia data are themselves a second-order solution on a  $129 \times 129$  grid. The order is therefore measured against a manufactured solution,

$$\psi(x, y, t) = g(t) \sin^2\left(\frac{\pi x}{L_x}\right) \sin^2\left(\frac{\pi y}{L_y}\right), \quad g(t) = 0.2 + 0.1 \sin(2\pi t). \quad (53)$$

It has  $\psi = 0$  and  $u = v = 0$  on every wall, so it uses the solver’s ordinary stationary no-slip walls and the  $\psi = 0$  Poisson condition without any change. It is made an exact solution by an optional source term  $f$  in the vorticity equation (the curl of a body force, were it written in the momentum equation),

$$\frac{\partial \omega}{\partial t} = -u \frac{\partial \omega}{\partial x} - v \frac{\partial \omega}{\partial y} + \frac{1}{\text{Re}} \nabla^2 \omega + f, \quad f = \frac{\partial \omega_e}{\partial t} + u_e \frac{\partial \omega_e}{\partial x} + v_e \frac{\partial \omega_e}{\partial y} - \frac{1}{\text{Re}} \nabla^2 \omega_e, \quad (54)$$

where the subscript  $e$  denotes the exact fields from eq. (53), all in closed form. The source is the `vorticity_source` callback of `solver_config`; it is unset in normal runs. The solver evaluates it at the time of each right-hand-side evaluation, so at  $t$ ,  $t + \Delta t/2$ ,  $t + \Delta t/2$  and  $t + \Delta t$  in RK4, with  $t = t_0 + n \Delta t$  after  $n$  steps of a run that starts at `t0`, and the CUDA backend fills it on the host and copies it to the device.

Each run starts from the exact state at  $t = 0$ , uses RK4 and the FFT solver at  $\text{Re} = 100$ , and is compared with the exact solution at  $t = 0.25$ . The time step is at most  $2.5 \times 10^{-3}$  and at most half the viscous stability limit; a run with a ten to thirty times smaller step gives the same errors to all digits shown, so the time error is negligible. The observed order between two grids is

$$p_{\text{obs}} = \log_2 \frac{\|e_h\|_\infty}{\|e_{h/2}\|_\infty}. \quad (55)$$

For derivative order 6 on the unit square:

| Grid    | $\psi$                       | $u$                          | $\omega$ , walls included    | $\omega$ , interior          |
|---------|------------------------------|------------------------------|------------------------------|------------------------------|
| $33^2$  | $1.02 \times 10^{-3}$        | $2.97 \times 10^{-3}$        | $2.89 \times 10^{-2}$        | $2.01 \times 10^{-2}$        |
| $65^2$  | $2.49 \times 10^{-4}$ (2.03) | $7.44 \times 10^{-4}$ (2.00) | $3.84 \times 10^{-3}$ (2.91) | $3.26 \times 10^{-3}$ (2.62) |
| $129^2$ | $6.12 \times 10^{-5}$ (2.02) | $1.86 \times 10^{-4}$ (2.00) | $6.19 \times 10^{-4}$ (2.63) | $5.50 \times 10^{-4}$ (2.57) |
| $257^2$ | $1.52 \times 10^{-5}$ (2.01) | $4.67 \times 10^{-5}$ (2.00) | $1.19 \times 10^{-4}$ (2.37) | $1.11 \times 10^{-4}$ (2.31) |
| $513^2$ | $3.78 \times 10^{-6}$ (2.01) | $1.17 \times 10^{-5}$ (2.00) | $2.58 \times 10^{-5}$ (2.21) | $2.47 \times 10^{-5}$ (2.16) |

(maximum errors over the nodes, observed order in parentheses). The findings:

- For this smooth solution with stationary no-slip walls, the complete method converges at second order in space with every derivative order.  $\psi$ ,  $u$  and  $v$  converge at 2.0 on every grid.  $\omega$  converges faster on the coarse grids and approaches second order under refinement; its largest error is at the walls.
- Orders 4 and 6 give the same errors to three digits, so the sixth-order rows of the deep interior do not affect the result. The order is held back by something all derivative orders share: the five-point Poisson operator (eq. (30)) and the low-order rows next to the walls (table 1). Section 14.6 separates the two.
- Compared with order 2, orders 4 and 6 reduce the vorticity error by about  $2.6\times$  at  $513^2$  ( $6.9 \times 10^{-5}$  to  $2.6 \times 10^{-5}$ , walls included) and leave the velocity error essentially unchanged ( $1.02 \times 10^{-5}$  for  $u$  with order 2,  $1.17 \times 10^{-5}$  with order 6).
- A  $2 \times 1$  domain and a unit-square grid with  $\Delta x = \Delta y/2$  show the same orders.

`make test` runs a short version: on  $17^2$ ,  $33^2$  and  $65^2$  it requires the observed orders of  $\psi$ ,  $u$  and  $v$  between  $33^2$  and  $65^2$  to be within 0.15 of 2 and that of  $\omega$  to be at least 1.9, and it bounds the errors at  $65^2$ . It also checks that Euler and RK4 keep their first and fourth order in time with the time-dependent source, which fails if an RK4 stage evaluates the source at the wrong time, and that a run started at  $t_0 = 0.4$  is as accurate as one started at  $t = 0$ . `make convergence` runs the full study up to  $257^2$  for orders 2, 4 and 6 and the two rectangular cases.

## 14.6 What limits the spatial order: an ablation

To find which part of the discretization holds the order to two, a copy of the RK4 step (with nothing replaced, it computes exactly what `step()` computes, bit for bit) replaces parts of the method with the exact solution at each stage's time:

- *exact wall*  $\omega$ : the wall vorticity is exact; the Poisson solve and the velocity  $u = D_y \psi$ ,  $v = -D_x \psi$  are as usual;

- *exact  $\psi$* :  $\psi$  is exact, so the Poisson operator plays no part;  $u$ ,  $v$  and the wall vorticity  $D_x v - D_y u$  are computed from it;
- *exact  $u$ ,  $v$* : the velocity is exact at every node; the wall vorticity is computed from it.

Combinations replace several parts. With exact  $u$ ,  $v$  and exact wall  $\omega$ , only the transport derivatives of  $\omega$  remain. For derivative order 6, maximum error of  $\omega$  over all nodes at  $257^2$ , with the observed order from  $129^2$ :

| Replaced by the exact solution | $\omega$ error       | order of $\omega$ | order of $u$ |
|--------------------------------|----------------------|-------------------|--------------|
| nothing (the solver)           | $1.2 \times 10^{-4}$ | 2.37              | 2.00         |
| wall $\omega$                  | $5.1 \times 10^{-5}$ | 2.00              | 2.00         |
| $\psi$                         | $1.2 \times 10^{-3}$ | 2.00              | 3.00         |
| $\psi$ and wall $\omega$       | $2.8 \times 10^{-7}$ | 4.02              | 3.00         |
| $u$ , $v$                      | $5.9 \times 10^{-4}$ | 2.00              | –            |
| $u$ , $v$ and wall $\omega$    | $2.7 \times 10^{-7}$ | 3.89              | –            |

Order 4 gives the same orders (measured up to  $129^2$ ); order 2 gives 2.00 in every variant.

- **Both the Poisson operator and the wall-vorticity closure limit the order to two, each on its own.** With the exact wall vorticity, the five-point Poisson operator still gives second order; with the exact  $\psi$ , the wall closure still gives second order. Replacing  $\psi$  by the exact solution also removes the coupling between  $\psi$  and the wall formula, which a real operator keeps; section 14.7 measures the real higher-order operators.
- **The two errors partly cancel.** Replacing only  $\psi$  by the exact solution makes the vorticity error ten times *larger* ( $1.2 \times 10^{-3}$  against  $1.2 \times 10^{-4}$ ), although the solver’s error contains the wall closure’s error as well as the Poisson operator’s: in the coupled scheme the two partly cancel. This is consistent with the observation that a locally low-order wall formula such as Thom’s can be part of a globally second-order method.
- **With both replaced, the order rises to four, not six.** With the exact velocity and wall vorticity, only the derivative operators are left, and  $\omega$  converges at 3.9, approaching four. For this solution, that is consistent with the operators’ rows next to the walls, which are of second and fourth order (table 1); the first- and second-derivative rows were not ablated separately. Without walls the same stencils reach sixth order (section 14.8).

`make ablation` prints this study for orders 2, 4 and 6 up to  $129^2$ ; `make test` checks three of its results on  $33^2$  and  $65^2$  (orders 2, 2 and 4 for exact wall  $\omega$ , exact  $\psi$ , and both).

## 14.7 Raising the order with walls

The compact Poisson operator (section 8.5) and the third-order wall closure (eq. (20)) replace the two parts the ablation identified. With derivative order 6, on the manufactured solution of section 14.5, maximum errors at  $257^2$  and the observed order from  $129^2$ :

| Poisson operator, wall closure       | $\omega$ error        | order of $\omega$ | order of $\psi$ | order of $u$ |
|--------------------------------------|-----------------------|-------------------|-----------------|--------------|
| five-point, velocity-based (default) | $1.19 \times 10^{-4}$ | 2.37              | 2.01            | 2.00         |
| five-point, eq. (20)                 | $9.65 \times 10^{-5}$ | 2.05              | 2.00            | 2.00         |
| compact, velocity-based              | $7.52 \times 10^{-5}$ | 3.01              | 2.99            | 2.79         |
| compact, eq. (20)                    | $3.16 \times 10^{-7}$ | 4.20              | 4.21            | 3.00         |
| and fourth-order velocity rows       | $3.06 \times 10^{-7}$ | 4.10              | 4.21            | 4.92         |

Derivative order 4 gives the same orders; order 2 stays at two.

- **Together they give fourth order.**  $\omega$  converges at 4.0 on the interior and at 4.2 including the walls,  $\psi$  at 4.2, and the error at  $257^2$  is 380 times smaller than the default's.
- **The wall closure alone changes nothing:** the five-point Poisson operator keeps the order at two.
- **The compact operator alone gives three,** more than the ablation's exact- $\psi$  variant suggested: with the real operator,  $\psi$  and the velocity-based wall formula stay coupled, and the result is one order above what the formula's local accuracy suggests, as is known for Thom's formula.
- **With these two, the velocity converges at three.**  $u = D_y\psi$  uses the first-derivative rows next to the walls (table 1), which limit it;  $\omega$  is not limited by them here.
- **Fourth-order velocity rows lift the velocity to fourth order.** With `-velocity-order 4` (`solver_config.DXv, DYv`),  $u = D_y\psi$  and  $v = -D_x\psi$  use a first-derivative operator whose rows at the wall and next to it are fourth-order one-sided,  $(-25f_0 + 48f_1 - 36f_2 + 16f_3 - 3f_4)/(12h)$  and  $(-3f_0 - 10f_1 + 18f_2 - 6f_3 + f_4)/(12h)$ , and whose other rows are those of `SDiff1` (`SDiff1_wall14`). Only the velocity and the continuity check use it, so the divergence of the velocity stays at round-off and the transport is unchanged. With the other two options,  $u$  and  $v$  then converge at fourth order: the new rows are formally fourth order, derivative order 4 measures 4.0, and derivative order 6 measures 4.9 over the finest grids for this solution, presumably a favourable error structure rather than a higher order; and the error of  $u$  at  $257^2$  falls from  $2.3 \times 10^{-6}$  to  $1.8 \times 10^{-9}$ . On its own, without the compact Poisson operator, it leaves the order at two.

All three options are stable at the same time step: the default cavity runs bounded for 3000 steps at 0.95 times the stability limit with RK4 and with Euler. The CPU and GPU solvers give the same result to  $10^{-15}$ . `make test` checks the compact operator's order, the pairing (order at least 3.7 from  $33^2$  to  $65^2$ , and two for the wall closure alone) and the GPU against the CPU; `make convergence` prints the four combinations.

**The cavity.** The corners of the lid make the cavity singular, so its order is not that of the manufactured solution, but the two options still converge much faster. For the default case ( $\text{Re} = 100$ ,  $t = 30$ ), the largest difference of the centerline velocities at the 15 interior Ghia points from the  $257^2$  solution with both options is  $3.2 \times 10^{-3}$  at  $64^2$  and  $5.6 \times 10^{-4}$  at  $129^2$  with the defaults, against  $4.8 \times 10^{-4}$  and  $1.3 \times 10^{-5}$  with both options ( $5.0 \times 10^{-4}$  and  $1.5 \times 10^{-5}$  with the velocity rows as well: the cavity's singular corners, not the velocity, limit it). The two  $257^2$  solutions agree to  $1.0 \times 10^{-4}$ , and both lie 0.005 ( $u$ ) and 0.009 ( $v$ ) from Ghia et al. Measured against Ghia et al., the  $64^2$  solution with both options looks worse than the default (0.0055/0.0090 against 0.0023/0.0073): the default's small distance at  $64^2$  comes partly from error cancellation (section 14.10), and the reference itself is a second-order solution on  $129^2$ . The defaults are unchanged; both options are opt-in.

## 14.8 Spatial order without walls

On the periodic grid of section 8.6, the manufactured solution is a sum of three Fourier modes,

$$\psi = g(t) \sum_{m=1}^3 A_m \cos\left(2\pi \left(\frac{a_m x}{L_x} + \frac{b_m y}{L_y}\right) + \varphi_m\right), \quad (56)$$

with  $(a_m, b_m) = (1, 1), (2, -1), (1, 3)$  and amplitudes  $(1, \frac{1}{2}, \frac{1}{4})/(2\pi)$ . The wavenumbers differ, so the nonlinear term does not vanish. With the same  $g(t)$ ,  $\text{Re}$ , final time and time-step control as in section 14.5, the maximum error of  $\omega$  is

| Grid    | order 2                      | order 4                      | order 6                      |
|---------|------------------------------|------------------------------|------------------------------|
| $32^2$  | $3.34 \times 10^{-1}$        | $1.87 \times 10^{-2}$        | $1.23 \times 10^{-3}$        |
| $64^2$  | $8.44 \times 10^{-2}$ (1.99) | $1.22 \times 10^{-3}$ (3.94) | $2.05 \times 10^{-5}$ (5.91) |
| $128^2$ | $2.11 \times 10^{-2}$ (2.00) | $7.68 \times 10^{-5}$ (3.98) | $3.25 \times 10^{-7}$ (5.98) |
| $256^2$ | $5.27 \times 10^{-3}$ (2.00) | $4.81 \times 10^{-6}$ (4.00) | $5.09 \times 10^{-9}$ (6.00) |

and  $\psi$ ,  $u$  and  $v$  show the same orders ( $\psi$  reaches  $1.9 \times 10^{-11}$  at  $256^2$  with order 6). Without walls, and with a Poisson operator of the same order, the method reaches the nominal order of its stencils, which completes the picture of section 14.6: the second order of the wall-bounded method comes from the wall closures and the five-point Poisson operator, not from the interior stencils. `make test` checks the orders between  $32^2$  and  $64^2$  (within 0.15 of 2, 4 and 6) and on a  $2 \times 1$  domain; `make regression` checks the Taylor–Green vortex, whose error falls by a factor of 64 from  $32^2$  to  $64^2$ .

## 14.9 Diagnostics

For the Taylor–Green vortex the integrals are  $E = \frac{1}{4}e^{-4\nu k^2 t}$ ,  $Z = 2k^2 E$  and  $P = 4k^4 E$  with  $k = 2\pi$ , and all energy lies in the shell  $|\mathbf{k}| = \sqrt{2}k$ . On  $32^2$  (order 6) the computed  $E$  and  $Z$  at  $t = 0$  are exact to round-off,  $P$  is within  $8 \times 10^{-7}$  (it uses  $D_x$ ,  $D_y$ ),  $E$  at  $t = 0.1$  is within  $6 \times 10^{-7}$  of the exact decay, and the spectrum holds all energy in that shell to  $10^{-14}$ . On the unforced three-mode periodic flow of section 14.8 at  $\text{Re} = 100$ :

| Order, grids        | energy transfer                             | enstrophy transfer                          | $ \dot{E} + 2\nu Z /2\nu Z$                 |
|---------------------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| 2, $64^2$ / $128^2$ | $1.2 \times 10^{-2}$ / $2.9 \times 10^{-3}$ | $1.3 \times 10^{-2}$ / $3.3 \times 10^{-3}$ | $1.1 \times 10^{-2}$ / $2.8 \times 10^{-3}$ |
| 4, $64^2$ / $128^2$ | $9.2 \times 10^{-4}$ / $5.9 \times 10^{-5}$ | $6.3 \times 10^{-4}$ / $4.0 \times 10^{-5}$ | $1.7 \times 10^{-4}$ / $1.1 \times 10^{-5}$ |
| 6, $64^2$ / $128^2$ | $4.2 \times 10^{-5}$ / $6.8 \times 10^{-7}$ | $2.5 \times 10^{-5}$ / $4.0 \times 10^{-7}$ | $3.0 \times 10^{-6}$ / $7.2 \times 10^{-8}$ |

(net nonlinear transfers, relative to the largest flux). The spectra sum to the integrals to round-off. The nonlinear term conserves energy and enstrophy, and the energy budget closes, up to errors that fall at the order of the derivatives. `make test` checks these, and the GPU integrals against the CPU; `make regression` checks the Taylor–Green energy written to `integrals.csv`.

## 14.10 Physical validation against Ghia et al.

The code embeds the standard  $\text{Re} = 100$  centerline velocity values from Ghia et al. [1] and writes them alongside the simulation profile. This supports a conventional cavity-flow validation workflow:

1. run to a demonstrably steady state;
2. interpolate the numerical centerline velocity to the Ghia sample coordinates, or vice versa;
3. compute  $L_2$ ,  $L_\infty$ , and selected pointwise errors;
4. repeat on successively refined grids;
5. repeat with smaller timesteps to separate spatial and temporal error.

For the default case ( $\text{Re} = 100$ , RK4, FFT solver,  $t_f = 30$ ), the simulated centerline velocities interpolated to the 15 interior Ghia points give:

| Grid                                       | $u$ on $x = 0.5$ |           | $v$ on $y = 0.5$ |           |
|--------------------------------------------|------------------|-----------|------------------|-----------|
|                                            | max error        | rms error | max error        | rms error |
| $64 \times 64$ , mixed convention (before) | 0.0594           | 0.0372    | 0.0595           | 0.0301    |
| $64 \times 64$ , nodal convention          | 0.0023           | 0.0014    | 0.0073           | 0.0039    |
| $65 \times 65$ , nodal convention          | 0.0026           | 0.0014    | 0.0076           | 0.0039    |
| $129 \times 129$ , nodal convention        | 0.0045           | 0.0024    | 0.0089           | 0.0049    |
| $257 \times 257$ , nodal convention        | 0.0050           | 0.0025    | 0.0091           | 0.0051    |

The solution has converged at these resolutions: at the Ghia points the  $129 \times 129$  and  $257 \times 257$  profiles differ by at most  $4.5 \times 10^{-4}$  ( $u$ ) and  $3.5 \times 10^{-4}$  ( $v$ ), an order of magnitude less than either differs from the reference. The remaining discrepancy is therefore not grid error of this code; it is of the order of the accuracy of the reference data themselves, which were computed on a  $129 \times 129$  grid with a second-order scheme. The errors grow slightly from  $64 \times 64$  to  $129 \times 129$ : the smaller error of the coarse grid comes partly from its discretisation error cancelling against the difference to the reference, so  $64 \times 64$  should not be read as the most accurate grid. The  $257 \times 257$  run used  $\Delta t = 3 \times 10^{-4}$ .

`make regression` automates this comparison for the default case: the steady centerline profiles must be within 0.004 ( $u$ ) and 0.010 ( $v$ ) of the reference values, which are stored with the tests rather than taken from the program’s output, and any value that is not a finite number fails. Grid-convergence rates are not measured on the cavity, for the reason given in section 14.5, but on a manufactured solution.

## 15 Performance Characteristics

The default FFT-based Poisson solver generally dominates the timestep cost, especially under RK4 where five elliptic solves are performed per step. Measured timings per timestep for the CPU, OpenMP and CUDA builds are kept in one place, the Performance section of `README.md`, so that they are not maintained in two copies; they are implementation benchmarks on one machine rather than performance guarantees.

For transform performance, grid sizes for which  $n-1$  factors into small primes are advantageous, because the sine transform of the  $n-2$  interior nodes acts through FFT lengths of  $2(n-1)$ . Sizes such as 65, 129, 257, 513 and 1025 are therefore fast, and their neighbours can be several times slower: on the machine of the README benchmarks one serial FFT Poisson solve takes 6.7 ms at  $255^2$  ( $2(n-1) = 508 = 4 \cdot 127$ ) against 1.6 ms at  $257^2$  (512), and 85 ms at  $1023^2$  against 31 ms at  $1025^2$ .

The GPU becomes increasingly advantageous as the grid grows because the fixed kernel-launch overhead is amortized over more work. Double-precision FFT throughput is a major cost on consumer GPUs, so larger speedups may be observed on hardware with stronger FP64 capability.

## 16 Building and Running

### 16.1 Dependencies

The CPU solver requires a C compiler, the math library, and FFTW3. OpenMP is optional. CUDA requires the NVIDIA CUDA toolkit and cuFFT.

On Ubuntu/Debian, a typical CPU setup is

```
sudo apt install build-essential libfftw3-dev
```

For CUDA development, the distribution toolkit or NVIDIA’s toolkit packages may be used.

## 16.2 Build modes

```
make                # serial CPU
make OPENMP=1       # OpenMP CPU
make CUDA=1         # CUDA-enabled binary
make OPENMP=1 CUDA=1 # CUDA plus OpenMP CPU fallback/tests
```

The Makefile records the active compiler flags so that changing OpenMP/CUDA modes triggers the required rebuild automatically.

## 16.3 Tests

```
make test
make CUDA=1 test
make OPENMP=1 CUDA=1 test
make OPENMP=1 convergence # spatial convergence study, about 1.5 minutes
```

The test program returns 0 when all executed checks pass, 1 when any check fails, and 77 when a CUDA test build cannot execute its GPU checks because no usable device is present.

Further checks, each a `make` target run by continuous integration: `test-cli` (invalid command lines are refused, with the program’s error status and message, without writing anything), `regression` (a short run against stored results and the steady default case against Ghia et al.), `test-asan` (AddressSanitizer and UndefinedBehaviorSanitizer), `valgrind` (no leaks, nothing still reachable), `format-check`, `cppcheck`, `tidy` (also over the CUDA and OpenMP branches of the C sources; the CUDA source itself is checked only by its compiler), and `WERROR=1` builds, which make `nvcc`’s warnings errors too. Continuous integration has no GPU, so the CPU–GPU comparisons run only where `make CUDA=1 test` is run on a GPU host.

## 16.4 Runtime options

The current binary accepts:

| Option                          | Description                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <code>-nx N</code>              | Number of grid points in $x$ .                                                                                                          |
| <code>-ny N</code>              | Number of grid points in $y$ .                                                                                                          |
| <code>-n N</code>               | Same number of grid points in $x$ and $y$ .                                                                                             |
| <code>-dt DT</code>             | Timestep.                                                                                                                               |
| <code>-tf TF</code>             | Final simulation time.                                                                                                                  |
| <code>-output-interval N</code> | Write VTK every $N$ steps; zero disables VTK output.                                                                                    |
| <code>-re RE</code>             | Reynolds number.                                                                                                                        |
| <code>-poisson-order N</code>   | 2 (five-point, default) or 4 (compact, section 8.5); FFT solver with walls.                                                             |
| <code>-wall-closure NAME</code> | <code>velocity</code> (default) or <code>briley</code> (eq. (20)).                                                                      |
| <code>-velocity-order N</code>  | 2 (default) or 4: fourth-order rows next to the walls for $u, v$ (section 14.7).                                                        |
| <code>-case NAME</code>         | <code>cavity</code> (default), or on a doubly periodic unit square <code>taylor-green</code> or <code>shear-layer</code> (section 8.6). |
| <code>-cpu</code>               | In a CUDA build, force the CPU backend.                                                                                                 |
| <code>-help</code>              | Print command-line help.                                                                                                                |

Example:

```
mkdir -p output
./cnavier --n 127 --dt 6.2e-4 --tf 20 --output-interval 50
```

Parameters such as Reynolds number, finite-difference order, Poisson method, and time integrator are presently configured in `src/main.c` rather than through command-line flags.

## 17 Default Numerical Configuration

At the repository state documented here, the source defaults are summarized in table 3.

Table 3: Principal defaults in `src/main.c`.

| Quantity                             | Default   | Meaning                          |
|--------------------------------------|-----------|----------------------------------|
| Re                                   | 100       | Reynolds number                  |
| $L_x, L_y$                           | 1, 1      | domain lengths                   |
| $n_x, n_y$                           | 64, 64    | computational array size         |
| $\Delta t$                           | 0.005     | timestep                         |
| $t_f$                                | 30        | final time                       |
| finite-difference <code>order</code> | 6         | sixth-order interior stencil     |
| <code>time\_scheme</code>            | 2         | classical RK4                    |
| <code>poisson\_type</code>           | 3         | FFT/DST-I direct discrete solver |
| Poisson iterate tolerance            | $10^{-3}$ | GS/SOR iterate-change threshold  |
| Poisson max iterations               | 10000     | GS/SOR iteration cap             |
| VTK output interval                  | 20        | steps between output files       |

## 18 Numerical Interpretation and Current Limitations

The following points are important when interpreting results scientifically.

### 18.1 Mixed formal spatial order

The transport derivatives may be sixth-order in the deep interior, but the boundary closures are lower order and the Poisson solve is based on a second-order discrete Laplacian. Measured against a smooth manufactured solution with stationary no-slip walls, the complete algorithm converges at second order in space for every derivative order, and orders 4 and 6 give the same errors (section 14.5). This is evidence for one solution family, not a proof for every flow. The ablation of section 14.6 shows that the Poisson operator and the wall-vorticity closure each limit the order to two on their own, so higher-order results need both improved; with the compact Poisson operator and the third-order wall closure (section 14.7) the method reaches fourth order for  $\omega$  and  $\psi$ , and with fourth-order velocity rows next to the walls for the velocity too. Beyond fourth order, the transport operators' rows next to the walls would have to be raised as well. On a doubly periodic grid, with no walls and a Poisson operator of the same order as the derivatives, the method reaches the nominal order 2, 4 or 6 (section 14.8).

### 18.2 Poisson operator differs from transport Laplacian

The diffusion term in eq. (4) uses the selected  $D_{xx} + D_{yy}$  operator, while the streamfunction equation always uses the standard five-point second-order Laplacian. This is a legitimate mixed discretization, but it should be described explicitly because the `order` setting does not uniformly change every spatial operator in the solver.

### 18.3 Boundary vorticity accuracy

Wall vorticity is reconstructed from differentiated wall velocities rather than from a dedicated high-order streamfunction wall formula. Its accuracy is therefore tied to the one-sided derivative closure and the array-edge interpretation.

### 18.4 Explicit stability

At startup a lid-speed Courant check and the viscous bound of the selected time integrator (computed from the assembled second-derivative operator, so it follows the grid, Reynolds number and finite-difference order) are enforced. These are stability bounds only; they say nothing about temporal accuracy, and users remain responsible for timestep-convergence studies.

### 18.5 Steady-state convergence

The default driver advances to a specified final time; it does not terminate based on a physical steady-state residual such as  $\|\omega^{n+1} - \omega^n\|$  or a momentum residual. For steady cavity studies,  $t_f$  should be selected after checking that monitored fields and centerline profiles no longer change materially.

### 18.6 Pressure is not solved

Pressure is absent from the vorticity–streamfunction formulation. This is an advantage for the cavity velocity field but means that pressure output is not currently a native solved quantity. If pressure is required, it must be reconstructed from an additional pressure-Poisson or momentum-based relation with suitable boundary conditions.

### 18.7 Geometry and dimensionality

The current method is two-dimensional and Cartesian. Curvilinear grids, immersed boundaries, unstructured meshes, three-dimensional vorticity stretching, and general inflow/outflow boundaries are outside the present scope.

## 19 Suggested Scientific Workflow

For a reproducible numerical experiment, the following workflow is recommended:

1. Record the Git commit, compiler, optimization flags, FFTW/CUDA versions, CPU/GPU, and runtime backend.
2. Run `make test` for the chosen CPU build; for CUDA work, also run the GPU comparison suite on the actual GPU host.
3. Select a baseline grid and timestep conservatively.
4. Confirm temporal convergence by reducing  $\Delta t$  while holding the grid fixed.
5. Confirm spatial convergence with at least three grids while keeping temporal error smaller than spatial error.
6. For the lid-driven cavity, compare centerline profiles against Ghia et al. and report interpolation/error norms.
7. For steady solutions, verify that extending  $t_f$  does not materially change the solution.
8. Report the discrete divergence extrema as an implementation consistency diagnostic, but do not substitute them for solution-convergence metrics.

## 20 Possible Methodological Extensions

The current architecture makes several research-oriented extensions relatively natural:

- a consistent fourth- or sixth-order Poisson discretization, solved by an appropriate sparse/direct/iterative method;
- dedicated high-order wall-vorticity formulas;
- adaptive timestep selection from advective and diffusive spectral bounds;
- automated steady-state residual monitoring;
- automated Ghia interpolation and error norms in the test suite;
- grid-convergence-index (GCI) reporting;
- alternative convection schemes, such as upwind-biased, compact, WENO, or skew-symmetric energy-conserving formulations;
- multigrid for the Poisson equation when boundary conditions or geometries no longer permit a separable sine transform;
- pressure reconstruction as a post-processing step;

## 21 Final Remarks

`cnavier` implements a mathematically clean and computationally compact route to incompressible flow in two dimensions: evolve vorticity, invert a scalar Poisson equation, and differentiate a streamfunction to recover velocity. The code is particularly transparent because the spatial derivatives are explicit finite-difference matrices, the two-dimensional operators are tensor products, and both CPU and GPU backends expose the same algorithmic sequence.

The strongest parts of the present methodology are the exact discrete streamfunction enforcement of incompressibility, the direct DST-based Poisson option, reusable sparse operators, RK4 stage consistency, and unusually thorough CPU–GPU equivalence tests for a compact CFD code. The main scientific qualifications are equally clear: high-order accuracy applies primarily to interior transport derivatives; boundary closures are lower order; the streamfunction Poisson operator is second order; and quantitative benchmark/grid-convergence validation is not yet automated.

For educational CFD, numerical experimentation, backend comparisons, and as a foundation for further research-oriented development, these characteristics make the project both useful and unusually auditable. For publication-quality quantitative claims, the next step is not to complicate the code immediately, but to formalize its grid convention and add systematic temporal/spatial convergence and benchmark-error measurements.

## A Mapping Between Equations and Source Code

| Mathematical operation     | Main implementation location                                                                    |
|----------------------------|-------------------------------------------------------------------------------------------------|
| First-derivative stencils  | <code>SDiff1()</code> in <code>src/finitediff.c</code>                                          |
| Second-derivative stencils | <code>SDiff2()</code> in <code>src/finitediff.c</code>                                          |
| Kronecker operators        | <code>skronecker()</code> in <code>src/linearalg.c</code> ; assembly in <code>src/main.c</code> |

| Mathematical operation    | Main implementation location                                                            |
|---------------------------|-----------------------------------------------------------------------------------------|
| CSR derivative evaluation | <code>spmv()</code> in <code>src/linearalg.c</code> ; <code>spmv\_kernel</code> on CUDA |
| Wall velocity condition   | <code>apply\_wall\_bc()</code> / <code>wall\_bc\_kernel</code>                          |
| Wall vorticity            | <code>step()</code> / <code>vorticity\_bc\_kernel</code>                                |
| Vorticity RHS             | <code>dwdt()</code> , <code>euler()</code> / <code>rhs\_kernel</code>                   |
| RK4                       | <code>rk4()</code> / CUDA stage kernels in <code>gpu\_step()</code>                     |
| Gauss–Seidel              | <code>poisson()</code> / <code>poisson\_redblack()</code>                               |
| SOR                       | <code>poisson\_SOR()</code> / <code>poisson\_redblack()</code>                          |
| FFT/DST-I Poisson         | <code>poisson\_FFT()</code> / <code>poisson\_fft()</code>                               |
| Velocity recovery         | <code>velocity\_from\_vorticity()</code> / <code>velocity\_kernel</code>                |
| Divergence diagnostic     | main-loop CSR operations / <code>continuity\_kernel</code>                              |
| Ghia centerlines          | <code>print\_centerline()</code> in <code>src/Utils.c</code>                            |
| Verification suite        | <code>tests/test\_solver.c</code>                                                       |

## B Reproducibility Record

This document was prepared from the public repository

<https://github.com/leo-aa88/cnavier>

with `main` observed at commit

5018e29bff534b252a7ac9025ece4ac43efd78ad

on 6 October 2026. Numerical details in this manual describe the source code at that repository state; later commits may change defaults or algorithms.

## References

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- [4] M. Frigo and S. G. Johnson, “The design and implementation of FFTW3,” *Proceedings of the IEEE*, vol. 93, no. 2, pp. 216–231, 2005.